

Multilingual Financial Fraud & Scam Detection (UPI + SMS)

Project Overview

This project is an AI-powered scam detection system designed specifically for India's **UPI and SMS ecosystem**. It analyzes incoming messages (including multilingual texts) and identifies potential financial fraud attempts such as phishing, fake KYC updates, OTP scams, and malicious links.

The system combines:

- **Machine Learning (Naive Bayes)** for classification
- **Keyword-based risk scoring**
- **Language detection & translation**
- **Entity extraction (UPI IDs, links, amounts)**

It provides a **clear risk level (LOW / MEDIUM / HIGH)** along with explanations and safety advice, making it useful for real-time fraud awareness.

Setup Instructions

1. Clone or Download Project

```
git clone <your-repo-link>
```

```
cd scam-detection
```

2. Create Virtual Environment (Recommended)

```
python -m venv venv
```

Activate:

- Windows:

```
venv\Scripts\activate
```

- Mac/Linux:

```
source venv/bin/activate
```

3. Install Dependencies

```
pip install -r requirements.txt
```

If requirements.txt is not available:

```
pip install pandas scikit-learn langdetect deep-translator colorama
```

Usage Guidelines

Run the Program

python main.py

Enter Message

You will be prompted:

Enter message:

Example inputs:

- Your account is blocked click here
- ₹5000 send to abc@upi urgent
- Kal milte hai class me (Hindi/mixed language)

What the System Does

1. Detects Language

- Supports multiple Indian languages (Hindi, Tamil, Telugu, Kannada, etc.)

2. Translates to English

- Ensures consistent analysis across languages

3. AI Prediction

- Classifies message as:
 - SAFE
 - SCAM

4. Risk Scoring

- Based on:
 - Suspicious keywords (OTP, KYC, urgent, etc.)
 - Links in message
 - Context signals

5. Entity Extraction

- Detects:

- UPI IDs
- URLs
- Monetary amounts

6. Provides Output

- Risk Level: LOW / MEDIUM / HIGH
- Confidence score
- Highlighted suspicious words
- Reasons for classification
- Safety advice

Sample Output (CLI)

SCAM DETECTION RESULT

 Message: Your account is blocked click here

 Language: English

 Translated: Your account is blocked click here

AI PREDICTION

Label: SCAM

Confidence: 0.95

RISK ANALYSIS

Level: HIGH

Score: 85

ENTITIES

UPI IDs: []

Links: []

Amounts: []

INSIGHTS

Highlighted: [ACCOUNT] is [BLOCKED] [CLICK] here

Reasons:

- Keyword detected: account
- Keyword detected: blocked
- Keyword detected: click

 Advice: Do NOT click links or send money.

Dependencies & Requirements

Python Version

- Python **3.8+** recommended

Libraries Used

- pandas → Data handling & logging
 - scikit-learn → Machine learning model (Naive Bayes)
 - langdetect → Language identification
 - deep-translator → Text translation
 - colorama → Colored terminal output
 - re (built-in) → Regex for entity extraction
 - uuid (built-in) → Unique request IDs
 - datetime (built-in) → Timestamping
 - os (built-in) → File handling
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Output File

- scam_logs.csv
 - Stores all analyzed messages with:
 - Timestamp
 - Message
 - Language
 - Risk level
 - AI prediction
 - Confidence score
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Key Features

-  Multilingual support (Indian languages)
 -  AI + Rule-based hybrid detection
 -  UPI, links & amount extraction
 -  Real-time CLI analysis
 -  Risk scoring with explanations
 -  Automatic logging for tracking scams
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Future Enhancements

- Integration with Android SMS APIs
 - Real-time UPI transaction monitoring
 - Deep learning (LSTM / Transformer models)
 - Web dashboard for visualization
 - API deployment for fintech apps
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